Chapter 3: Deep Learning

Deep Learning is a part of machine learning that uses neural networks with multiple layers. Neural networks contain interconnected units that transform input information into useful representations. During training, the network adjusts its parameters to reduce the difference between its predictions and the desired outputs.

Convolutional Neural Networks are widely used for image-related tasks, while recurrent and transformer-based architectures can process sequences and language. Deep learning often requires substantial data and computational resources. Important concepts include training, validation, loss functions, optimization, epochs, batch size, and overfitting.